BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

BOTANY

SECTION 'A' (Multiple Choice Questions) (22)

Maximum Marks: 22

Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Board typos kept verbatim (e.g. 'Anglosperms','Phtosynthetic','Lysosormes','Extermely','Cycnds','Phagocytonsis'). Scan check-marks/ underlines on correct options are not transcribed.)

Note: TIME: 1 Hour 30 Minutes (45 Marks) — AS PER CONDENSED SYLLABUS

Q1. Choose the correct answer for each from the given options:

Qi. Gametophytic generation is dominant in:

(A) Anglosperms
(B) Bryophytes
(C) Pteridophytes
(D) Gymnosperms

Qii. Oxygen becomes attached to RUBP during:

(A) Cs Pathway
(B) C4 Pathway
(C) Calvin-Benson
(D) Photorespiration

Qiii. This disease is caused by arbovirus:

(A) Mumps
(B) Yellow fever
(C) Measles
(D) Rabies

Qiv. He proposed the five kingdom system:

(A) G.I Nicolson
(B) Robert Whittaker
(C) M. Scheleiden
(D) Carl Benda

Qv. Nostoc fixes Nitrogen is:

(A) Heterocyst
(B) Akinetes
(C) Hormogonis
(D) Vegetative cells

Qvi. It has its own DNA:

(A) Vacuole
(B) Mitochondria
(C) Lysosormes
(D) Endoplasmic Reticulum

Qvii. Phtosynthetic bacteria liberate it:

(A) O2
(B) CO2
(C) H2S
(D) Sulphur

Qviii. It is the only living genus of sphenopsida:

(A) Rhynia
(B) Equisetum
(C) Cycnds
(D) Pinus

Qix. It is the self-digestion process:

(A) Heterophagy
(B) Autophagy
(C) Phagocytonsis
(D) Pinocytosis

Qx. The transportation required energy is:

(A) Active transport
(B) Passive transport
(C) Diffusion
(D) Phagocytosis

Qxi. Cristae is the part of:

(A) Golgi apparatus
(B) Vacuole
(C) Chloroplast
(D) Mitochondria

Qxii. Measles and mumps in human beings are caused by:

(A) Rhino virus
(B) Arbovirus
(C) Para myxo virus
(D) Rhabdo virus

Qxiii. Hepatitis type which passes through blood from mother to child during pregnancy is:

(A) Hepatitis A
(B) Hepatitis B
(C) Hepatitis C
(D) Hepatitis D

Qxiv. Extermely thin appendages helping during conjugation in bacteria are:

(A) Flagella
(B) Pilli
(C) celia
(D) Tentacles

Qxv. Bacterial growth will be faster during this stage:

(A) lag
(B) log
(C) stationary
(D) decline

Qxvi. Protoctist having Isomorphic alternation of generation:

(A) chlorella
(B) euglena
(C) ulva
(D) ulothrix

Qxvii. Most serious plant diseases caused by this class of Fungi:

(A) Ascomycota
(B) Basidiomycota
(C) Zygomycota
(D) Deuteromycota

Qxviii. Stigma is the part of:

(A) sepal
(B) petal
(C) stamen
(D) carpel

Qxix. Ascocarp having small opening at the apex is called:

(A) Apothecium
(B) Perithecium
(C) Cleistothecium
(D) Protothecium

Qxx. Process which converts pyruvate into three molecules of CO2 is called:

(A) C3 Pathway
(B) Kreb's cycle
(C) C4 cycle
(D) Glycolysis

Qxxi. The basic unit of biological classification is:

(A) Division
(B) Order
(C) Species
(D) Class

Qxxii. During Aerobic respiration complete oxidation of one Glucose molecule produces these many ATPs:

(A) 42
(B) 34
(C) 38
(D) 30

SECTION 'B' (Short-Answer Questions) (14)

Maximum Marks: 14

Q2.

Attempt any SIX part questions. Each question carries one mark. Answer should not exceed two lines: i) Why is ATP called Biological currency of energy? ii) Name any two diseases caused by Bacteria in human beings iii) Define Heterospory. iv) Name three methods of genetic recombination found in Bacteria. v) Why is Protoctista considered a Polyphyletic

kingdom? vi) Name only the sub-divisions of Tracheophyta. vii) Why is Anthocerotae considered an advance class of Bryophyta? viii) What is chemosynthetic nutrition? ix) Define Imbibition OR Osmosis.

Q3.

Attempt any four part questions. Each question carries two marks. Answer should not exceed five lines: i) Draw well labelled diagram of any one of the following: L.S of Ovule / T.S of Marchantia Thallus ii) Why are Bryphytes called the amphibious plants. iii) Discuss the steps involved in evolution of seed. iv) Write any four features of Cyanobacteria. v) Distinguish between any one of the following: Angiosperm and Gymnosperm / Alcoholic Fermentation & lactic acid fermentation vi) Classify Bacteria on the basis of flagella. vii) Mention any two factors which affect the opening and closing of stomata.

SECTION 'C' (Detailed Answer Questions) (09)

Maximum Marks: 09

Note: Attempt any Two questions from this section. All questions carry equal marks.

Q4.

Describe the life cycle of fern. (Diagram no required)

Q5.

Draw and describe the life cycle of Zygomycota.

Q6.

Describe the light dependent reactions of photosynthesis.

Q7.

Discuss Transpiration pull theory.

ZOOLOGY

SECTION 'A' (Multiple Choice Questions) (20)

Maximum Marks: 20

Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Board typos kept verbatim (e.g. 'Paeudopodia','Fiagella','coeloem','Rheuses Monkey'). Q(i) question text obscured in scan — marked. Scan check-marks/underlines on correct options are not transcribed.)

Note: TIME: 1 Hour 30 Minutes (40 Marks) — AS PER CONDENSED SYLLABUS

Q1. Choose the correct answer for each from the given options:

Qi. [question text obscured in source scan] Choose the correct answer for each from the given options:

(A) DNA
(B) RNA
(C) Chromosome
(D) Nucleosome

Qii. Condensation involves:

(A) Hydration
(B) Dehydration
(C) Isomerization
(D) Filtration

Qiii. The percentage of co-enzyme in an enzyme is about:

(A) 50
(B) 1
(C) 2
(D) 10

Qiv. Locomotion in Euglena takes place by:

(A) Cilia
(B) Paeudopodia
(C) Fiagella
(D) Myonemes

Qv. Presence of secondary and tertiary chambers is the characteristic of this system:

(A) Ascon
(B) Sycon
(C) Leucon
(D) Simple Sycon

Qvi. The phenomenon of alternation of generation is found in this given phylum:

(A) Porifera
(B) Plathelminthes
(C) Cnidaria
(D) Arthropoda

Qvii. The order of lung fish is:

(A) Apoda
(B) Dipnoi
(C) Myxiniformes
(D) Batodei

Qviii. Total number of canine in an adult person is:

(A) 4
(B) 8
(C) 12
(D) 6

Qix. The number of spiracles in cockroach is:

(A) 10 pairs
(B) 8 pairs
(C) 6 pairs
(D) 20 pairs

Qx. The heart of fish consists of:

(A) 2 chambers
(B) 3 chambers
(C) 4 chambers
(D) 1 chamber

Qxi. The conduction of impulse from S.A code to the A.V node in the heart requires:

(A) 0.5 second
(B) 1 second
(C) 0.15 second
(D) 0.25 second

Qxii. This immunity is involved in the treatment of rabies in man:

(A) Natural Active Immunity
(B) Natural Passive Immunity
(C) Artificial Active Immunity
(D) Artificial Passive Immunity

Qxiii. The coeloem of body cavity is formed due to the splitting of:

(A) Ectoderm
(B) Mesoderm
(C) Endoderm
(D) Mesoglea

Qxiv. The respiratory pigment is absent in:

(A) Earth worm
(B) Frog
(C) Rabbit
(D) Cockroach

Qxv. This molecule binds to the active site of an enzyme:

(A) Allosteric activator
(B) Allosteric inhibitor
(C) Non-competitive inhibitor
(D) Competitive inhibitor

Qxvi. Malaria is caused by:

(A) Amoeba
(B) Bacteria
(C) Virus
(D) Plasmodium

Qxvii. 1.4 pH is best for:

(A) Pepsin
(B) Ptyalin
(C) Amylase
(D) Lipase

Qxviii. Trypanosoma belongs to:

(A) Flagella
(B) Sarcodina
(C) Ciliata
(D) Suetoria

Qxix. This is not a fish:

(A) Eel fish
(B) Cat fish
(C) Dog fish
(D) Cuttle fish

Qxx. The national mammal of Pakistan is:

(A) Rabbit

(B) Markhor

(C) Camel

(D) Rheuses Monkey

SECTION 'B' (Short-Answer Questions) (12)

Maximum Marks: 12

Q2.

(a) Reasoning Questions: Attempt Six part questions. Select three from Reasoning Questions and Three from Non-reasoning questions: (i) How does substrate concentration affect the activity of enzymes? (ii) Why are amphibians successful land vertebrates? (iii) Why is the liver known as a metabolic factory in the body? (iv) Why is a double circuit heart in animals better than a single circuit heart? (v) Why is air better respiratory medium than water? (vi) Why is Archaeopteryx considered as connecting link between Reptiles and Birds?

(b) Non-Reasoning Questions: vii) Define Population and Community. viii) Differentiate between Chondriethyes and Osteichthyes. ix) Draw a neat and labelled diagram of Inspiration and Expiration in Man (ventral view). ix) (OR) Write the conditions necessary for a respiratory surface in animals. x) Write the functions of lymphatic system. xi) Write the parasitic adaptations in Helminthes. x) Write note on condensation.

SECTION "C" (Detailed Answer Questions) (09)

Maximum Marks: 09

> Note: Attempt any One question from this section. All questions carry equal marks. | نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

Q3.

a) Describe the salient features of phylum Annelida OR mollusca. Classify the phylum and give the characteristics of each class.

b) Write the characteristics of any one class that belongs to Amniotes.

Q4.

a) Describe the digestive system of man (Diagram is not required)

a) (OR) Explain Adaptive Immune System.

b) Write note on different types of mouth parts in insects.

Q5.

a) Describe the internal structure of the human heart and draw its neat and labelled diagram.

b) Differentiate between Artery and Vein.